

Artificial intelligence enabled ChatGPT and large language models in drug target discovery, drug discovery, and development

Drug discovery has been revolutionized recently by different computational approaches like artificial intelligence (AI), deep learning (DL), and quantum mechanical methods. Drug discovery and development was a prolonged process previously. Developing one small-molecular drug might take 15 years on average, costing around US \$2 billion. A paradigm shift has recently been noted in drug discovery using computational technologies during the past few years.^{1,2} Biotech-based drug discovery companies developed their new business models guided by molecular modeling techniques with AI and DL for “small-molecule hit generation” and “hit to lead generation.”^{3,4} In this direction, computational-guided ultra-large physics-based virtual screening techniques have been used in drug discovery.² All these computational approaches reduce the cost and time of early drug discovery stages. Therefore, biotech companies rely entirely on different computational approaches, especially AI, to search for new drugs. These companies look at this picky time for new algorithms to boost their drug discovery businesses. In this scenario, AI-enabled ChatGPT or large language models (LLMs) are the new addition to the drug discovery landscape.

During drug discovery, researchers applied several methods, such as virtual library design, high-throughput screening, and structure-based drug design, to identify small molecules. Structure-based drug design is a significant drug discovery technique that provides us with a better understanding of the interaction of therapeutic molecules and the drug target.^{5–7} In this area, potential drug target discovery is the first step that is essential to trigger the drug discovery process of different diseases.⁸ For potential drug target discovery, recent computational approaches and the “omics” perspective helps in the drug target discovery initiative.⁹ ChatGPT or LLMs are also helping in the drug target discovery process. Savage explained in a news article in *Nature Biotechnology* that ChatGPT or LLMs help the drug target discovery process. ChatGPT provides initial significant information in a drug target discovery of a specific disease, which may be validated subsequently. Savage illustrated that ChatGPT v.4 has a plug-in that can be helpful for the further process of drug discovery.¹⁰ Such an AI-enabled platform can be used for initial understanding to protein-based drug target discovery. Such ChatGPT derived information will help us to understand the structure of proteins, the structural domain in the protein, the drug-binding pocket, the active site of the protein-based drug target, etc. (Figure 1A).

Recently, Zhao and Wu have described how ChatGPT can accelerate the drug discovery process to execute the idea of innovative drugs. They discussed how this LLM can provide the blueprint of drug compounds with new structures. It also helps to predict the different features of a drug, such as pharmacodynamics (PD), pharmacokinetics

(PK), and toxicity of the molecule (Figure 1B).¹¹ Such information using ChatGPT will be helpful for pharmaceutical companies for drug discovery and development. These ChatGPT-derived pieces of information will trigger the drug discovery and development process to move more quickly. At the same time, it will also help the pharmaceutical industry to develop more drugs that will benefit patients.

Sharma and Thakur have illustrated that ChatGPT can assist in the process of drug discovery, such as computing the assortment of a compound, converting smiles to structure, creating input files for Gaussian software, generating input files for docking, finding PDB files, etc. At the same time, ChatGPT can assist in understanding the absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties of a drug. Similarly, it helps with optimizing drug properties and assessing drug toxicity. This tool is significant for early-stage drug discovery, although experimental validation is required.¹²

Serious consequences have been found from drug-drug interactions (DDIs) in patients. Therefore, DDI study is one of the significant parameters in the drug discovery and development process. It is being noted in the well-being of the patients. Recently, Juhi et al. explored the capacity of ChatGPT to predict and explain DDIs. The researchers analyzed 40 pairs of previously listed DDIs using ChatGPT with a two-stage question. In the first question, they noted that one answer was incorrect among the 40 DDI pairs. They also found that 19 were conclusive and 20 were inconclusive among the correct answers. The researchers concluded that ChatGPT might be an efficient tool for evaluating DDIs. Patients should use the tool to obtain initial information about DDIs.¹³

Presently, drug repurposing is one significant area for drug development. During the COVID-19 pandemic, several drugs were repurposed against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).^{14–16} Since then, drug repurposing has been an important technique for drug discovery and development. Recently, BioXcel Therapeutics, a US-based biopharmaceutical company, has been trying to develop immuno-oncology and neuroscience medicines. Recently, BioXcel Therapeutics has been trying to identify therapeutics for repurposed drugs using AI. However, the company is considering LLMs to pick the potential drug candidates.¹⁰

Several people have talked about domain-specific LLMs or next-generation ChatGPT, which should be error free and provide users with correct information.¹⁷ Recently, Liang et al. developed a pharmaceutical domain-specific LLM named “DrugChat.” It is an LLM that can analyze the drug compound and has ChatGPT-like properties to furnish

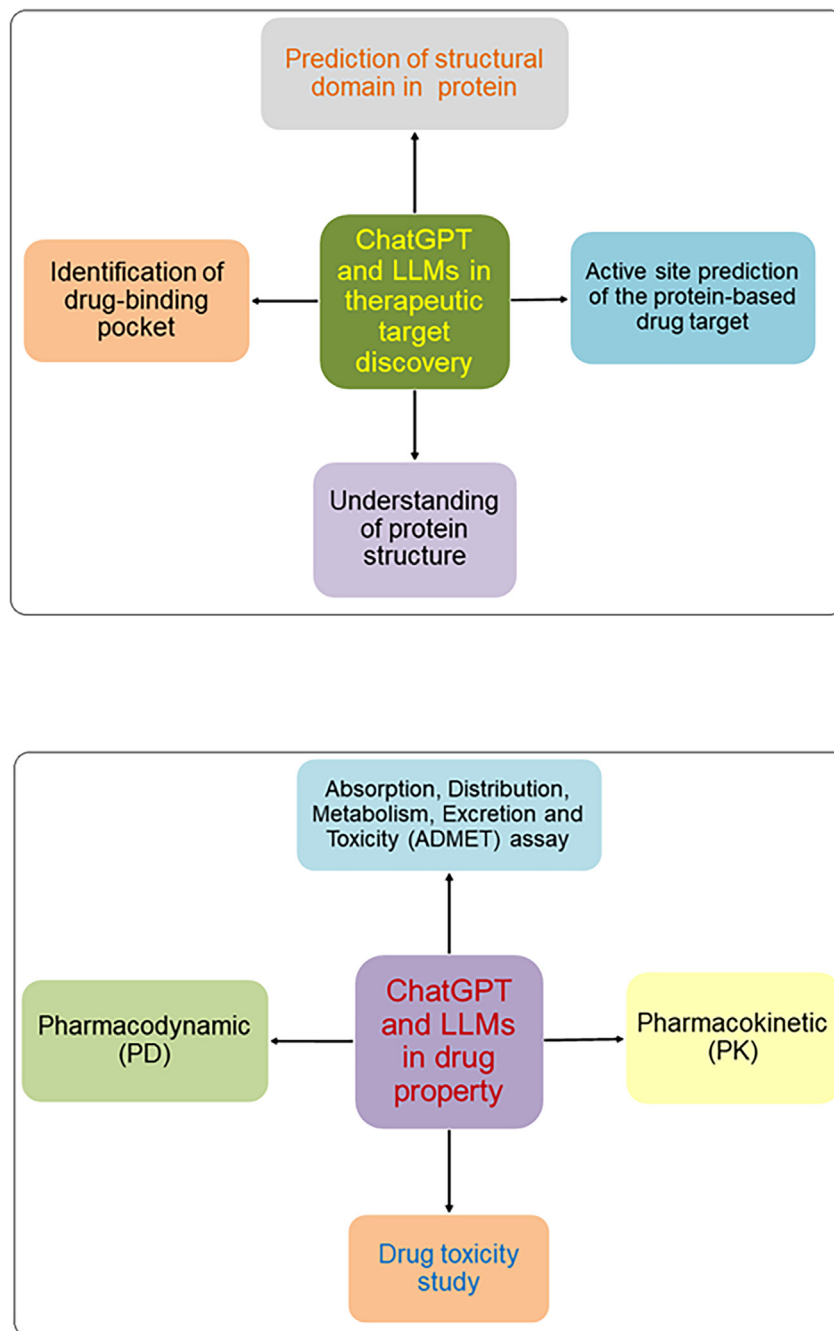


Figure 1. The role of ChatGPT and large language models (LLMs) in drug target discovery, drug discovery, and development

(A) The role of ChatGPT and LLMs in different processes of drug target discovery, such as protein domain prediction, drug-binding pocket identification, active site prediction, and structure prediction.

(B) The role of ChatGPT and LLMs in different drug property measurements, such as PD, PK, ADMET, toxicity, etc.

information on drug molecule graphs. The tool works similarly to ChatGPT. It comprises three main components: an LLM, a graph neural network (GNN), and an adaptor. DrugChat is trained by a vast number of drug datasets, which contain 10,834 drug compounds.¹⁸

Several other vital analyses using LLMs might help drug discovery research, such as genetic and protein structure data representation in text form. Recently, genetic and protein structure data are represented in text form and are amenable to the natural language

processing (NLP) technique that LLMs facilitated. Using DL and LLMs, Madani et al. illustrated a special LLM named “ProGen,” which can generate different protein sequences. This specialized LLM was trained on 280 million protein sequences from diverse protein families (about 19,000).^{19,20} AI-based research can provide information about the cell platform and its molecules. Recently, Heck illustrated that the AI-based tool could provide 70 kDa heat-shock protein information. It might be one of the applications of ChatGPT’s that helps in protein drug design.²¹

ChatGPT’s entrance into the research field will accelerate drug discovery research development. Different pharmaceutical and biotechnology companies are already using ChatGPT and LLMs in new therapeutic molecule discoveries, and this will finally benefit patients soon.

DATA AND CODE AVAILABILITY

The authors confirm that the data supporting the findings of this study are available within the article.

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